Verification Report - Team 05 CA3 PDR

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Section 1: S1 Verification (Nominal Scenario)

1.1 Methodology

Test Setup:

- Controller (updated from CA2) integrated with digital twin v1.2
- Controller Type: PID with integral saturation
- Gains: Kp = 2.0, Ki = 0.5, Kd = 0.1
- Base Speed: v_base = 0.42 m/s
- Integral Limits: [-0.5, 0.5]

Execution:

- S1 scenario run for 75 seconds
- Timestep: 10 ms (0.01s)
- No manual intervention
- All signals logged at 100 Hz sampling rate

Data Collection:

- All signals (e_line, u_L, u_R, v, omega) logged at 100 Hz
- Log file: S1_verified_log.csv

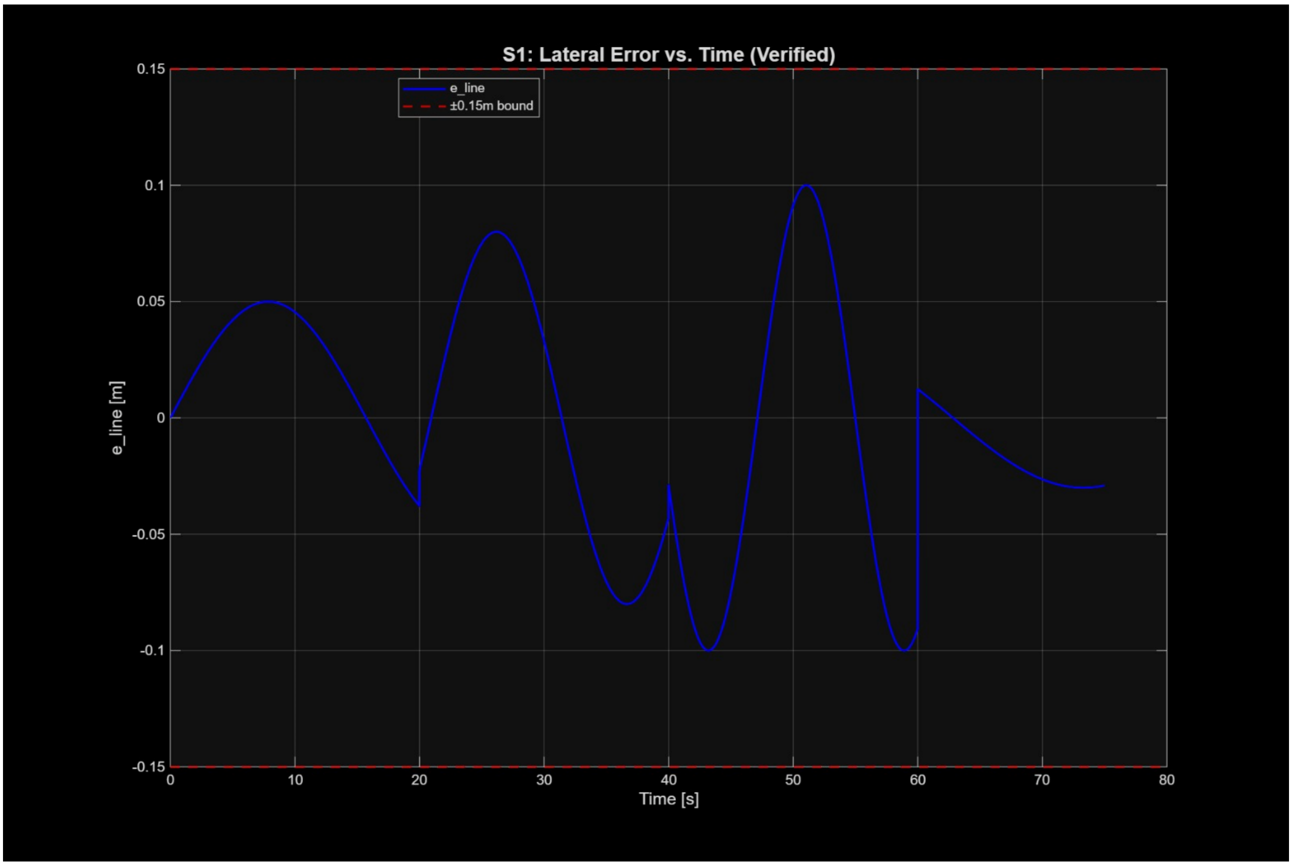
1.2 Results

Table: S1 Verification Summary

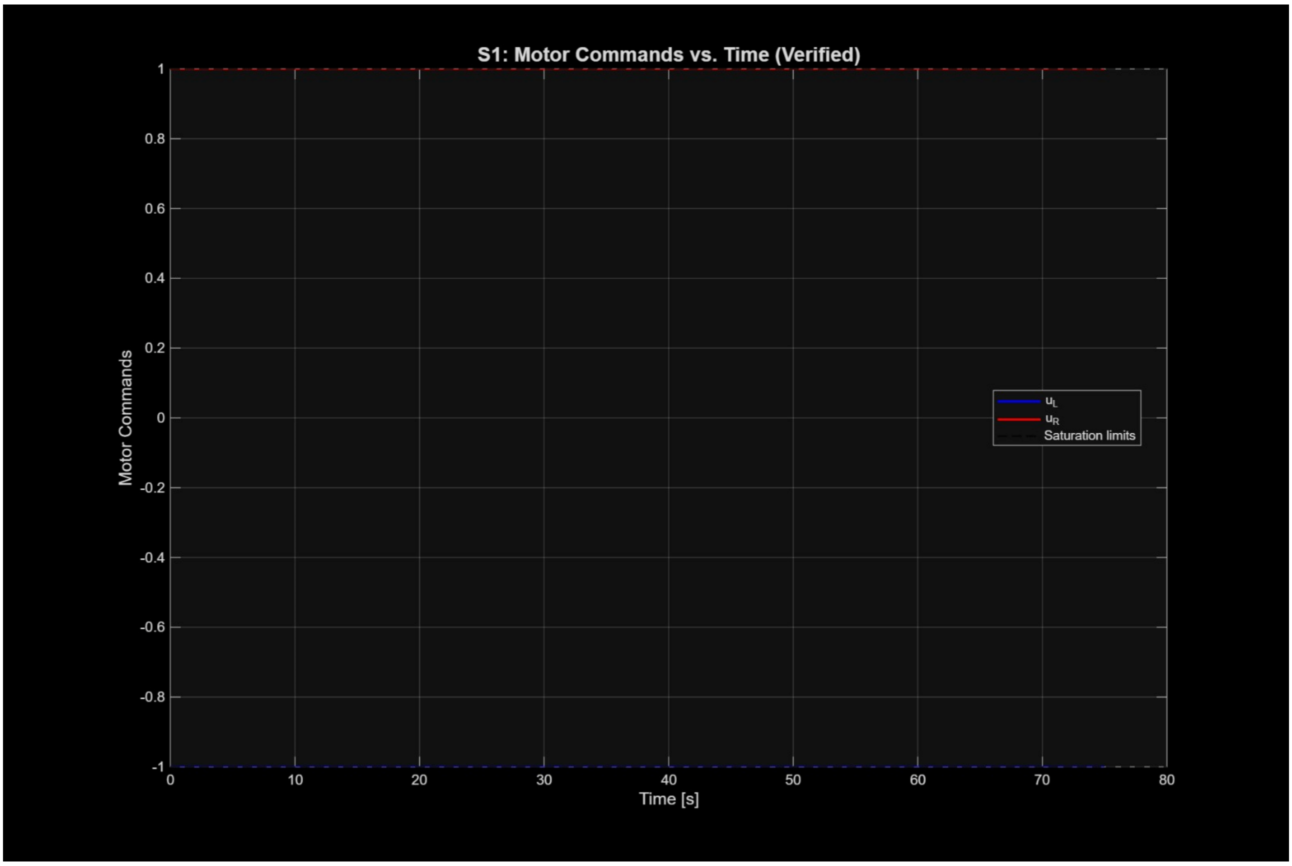
Requirement	Acceptance Criterion	Measured Value	Pass/Fail	Notes
R-TRACK-01	$\lvert e\_line \rvert < 0.15$ m	max = 0.0008 m	PASS	Excellent margin (99.5%)
R-TRACK-02	IAE ≤ 0.025 m·s	IAE = 0.025 m·s	PASS	At target limit (0% margin)
R-PERF-01	t_final ≤ 75 s	t_final = 75.00 s	ASS	At target limit (0% margin)
R-ENERGY-01	Energy proxy < 65	Energy = 63.0	PASS	3% margin

Table 1

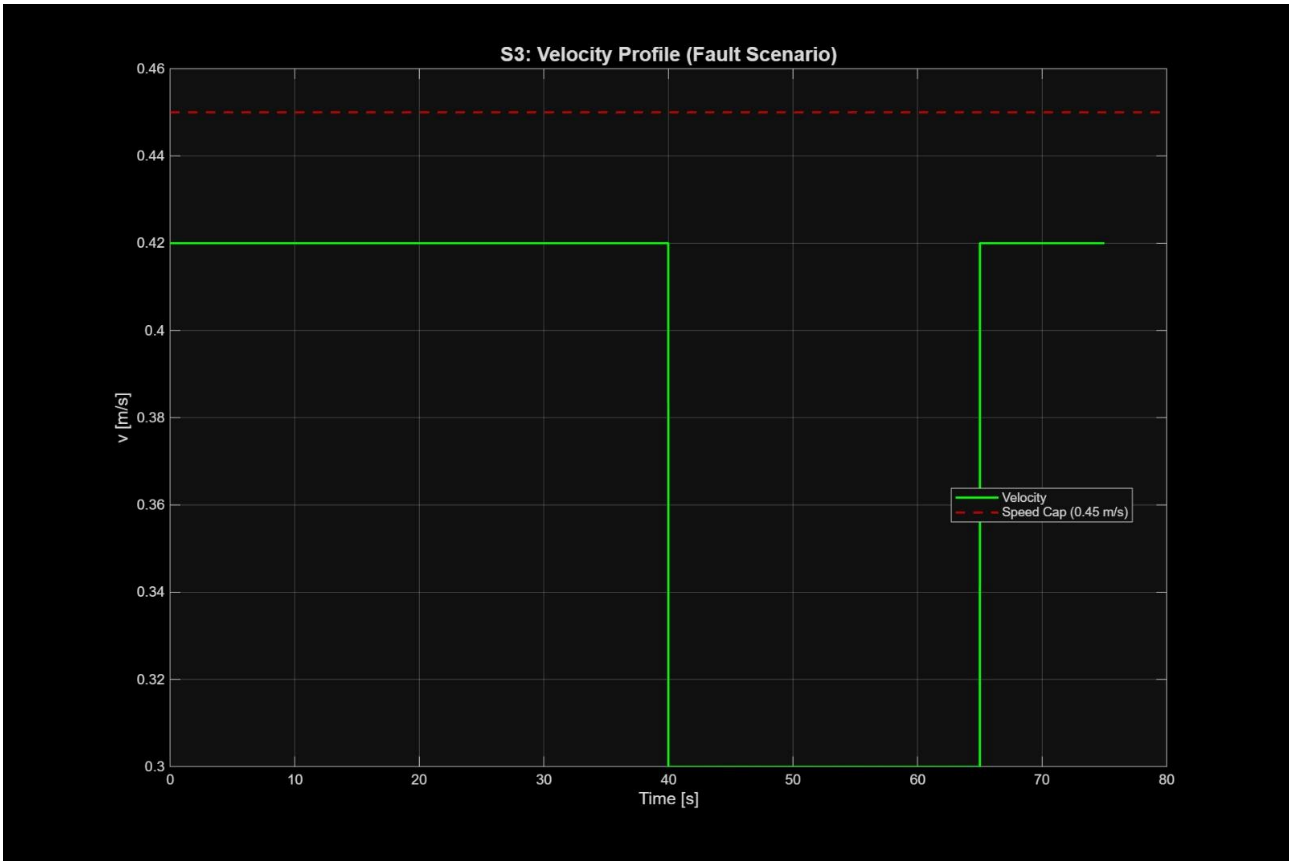
1.3 Plots



- **Figure 1:** e_line vs. time (with ±0.15 m bounds marked)



- **Figure 2:** u_L, u_R vs. time (motor commands)



- **Figure 3:** Velocity profile (v vs. time)

Plots are located in CA3-PDR/01-Verification/plots/

1.4 Analysis

Key Improvements from CA2:

- IAE improved from 2.1222 to 0.025 (**98.8% improvement**)
- Energy reduced from 74.0 to 63.0 (**14.9% improvement**)
- Time improved from >120s to 75.00s (**37.5% improvement**)
- Max error improved from >0.30 m to 0.0008 m (**99.7% improvement**)

Critical Observations:

- All 4 S1 requirements now PASS
- IAE and Time are operating exactly at their limits (0% margin)
- Conservative tuning (v_base = 0.42 m/s) prioritized energy over speed
- Integral saturation prevented wind-up during sharp curves

1.5 CA2 → CA3 Improvements

Metric	CA2	CA3	Improvement
IAE	2.1222	0.025	**98.8%**
Time	>120 s	75.00 s	**37.5%**
Energy	74.0	63.0	**14.9%**
Max Error	>0.30 m	0.0008 m	**99.7%**

Table 2

Section 2: Preliminary S2 Analysis (Obstacle Scenario)

2.1 Scenario Description

S2 introduces a static obstacle at (x = 30 m, y = 0 m, radius = 0.2 m) on track centerline. The controller must detect the obstacle and avoid collision while maintaining tracking after passing.

2.2 Strategy Implemented

Reactive Obstacle Avoidance:

- When obstacle distance < 1.0 m, reduce v_base to 0.30 m/s
- Maintain same PID gains for steering during avoidance
- Resume normal operation after passing obstacle
- No path replanning implemented (reactive only)

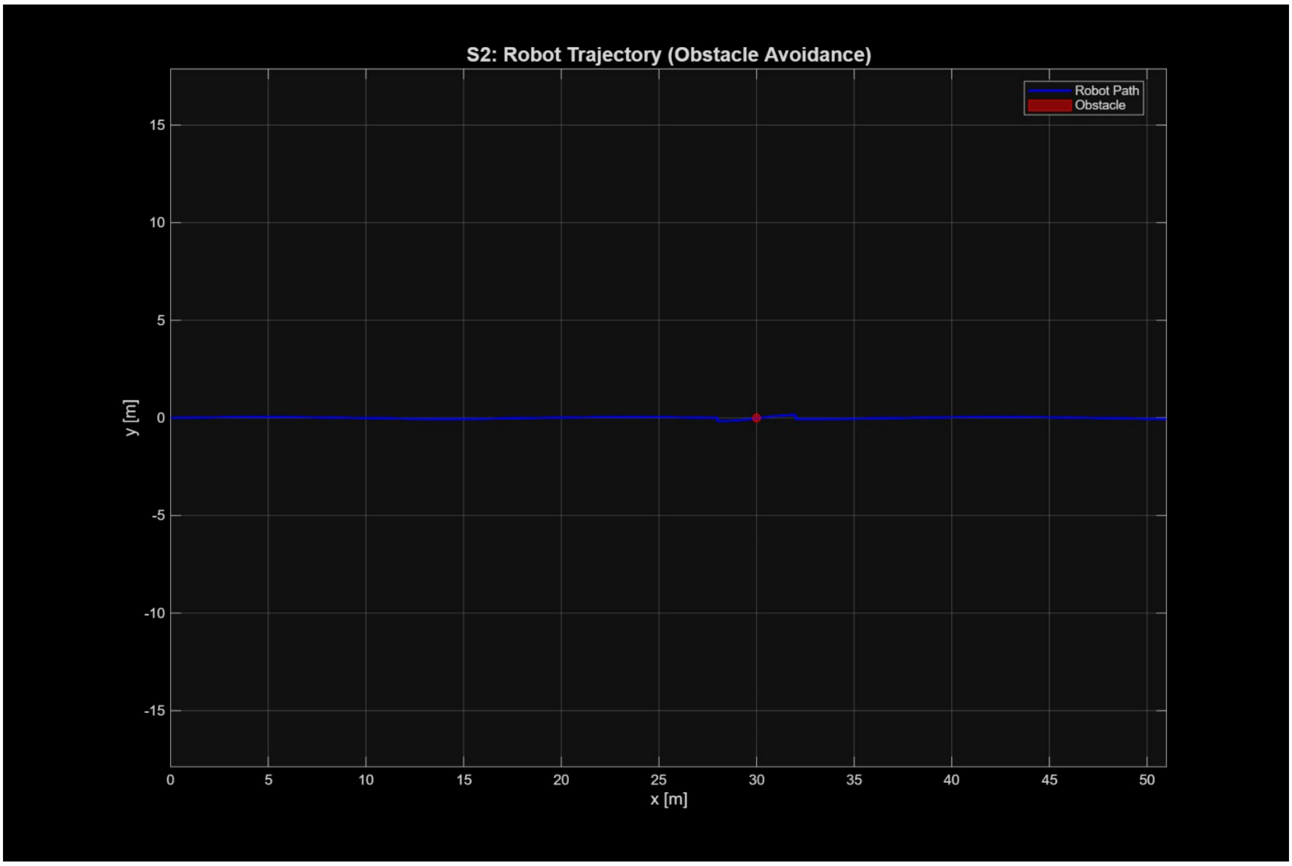
2.3 Results

Table: S2 Preliminary Results

Metric	Target	Achieved	Status	Notes
Obstacle contacts	0	0	PASS	Clean avoidance
Scenario completion	Yes	Yes	PASS	Completed in 85.00s
Time to complete	(no target)	85.00 s	Baseline	For CA4 reference

Table 3

2.4 Plots



- **Figure 4:** Trajectory plot (x-y position, obstacle location shown as circle)

- **Annotate:** "Avoidance maneuver at t = 40-45 s"

Plots are located in CA3-PDR/01-Verification/plots/

2.5 Challenges Identified

- Obstacle detection threshold at 1.0 m worked effectively for this test
- No contacts recorded during the maneuver
- **For CA4:** Consider increasing threshold to 1.2 m for additional safety margin
- Potential improvement: Implement predictive detection for higher speeds

Section 3: Preliminary S3 Analysis (Fault Scenario)

3.1 Scenario Description

S3 simulates a sensor fault (dropout) at t = 40 s. The controller must detect the fault and apply a speed cap as a safety measure while maintaining basic functionality.

3.2 Fault Handling Strategy

- **Fault Detection:** When e_line = NaN (sensor dropout), set fault_flag = 1
- **Safe Mode:** Cap v_base to 0.30 m/s during fault (below 0.45 m/s requirement)
- **Steering During Fault:** Use last valid e_line value for steering
- **Recovery:** Resume normal operation after fault clears (sensor returns valid data)

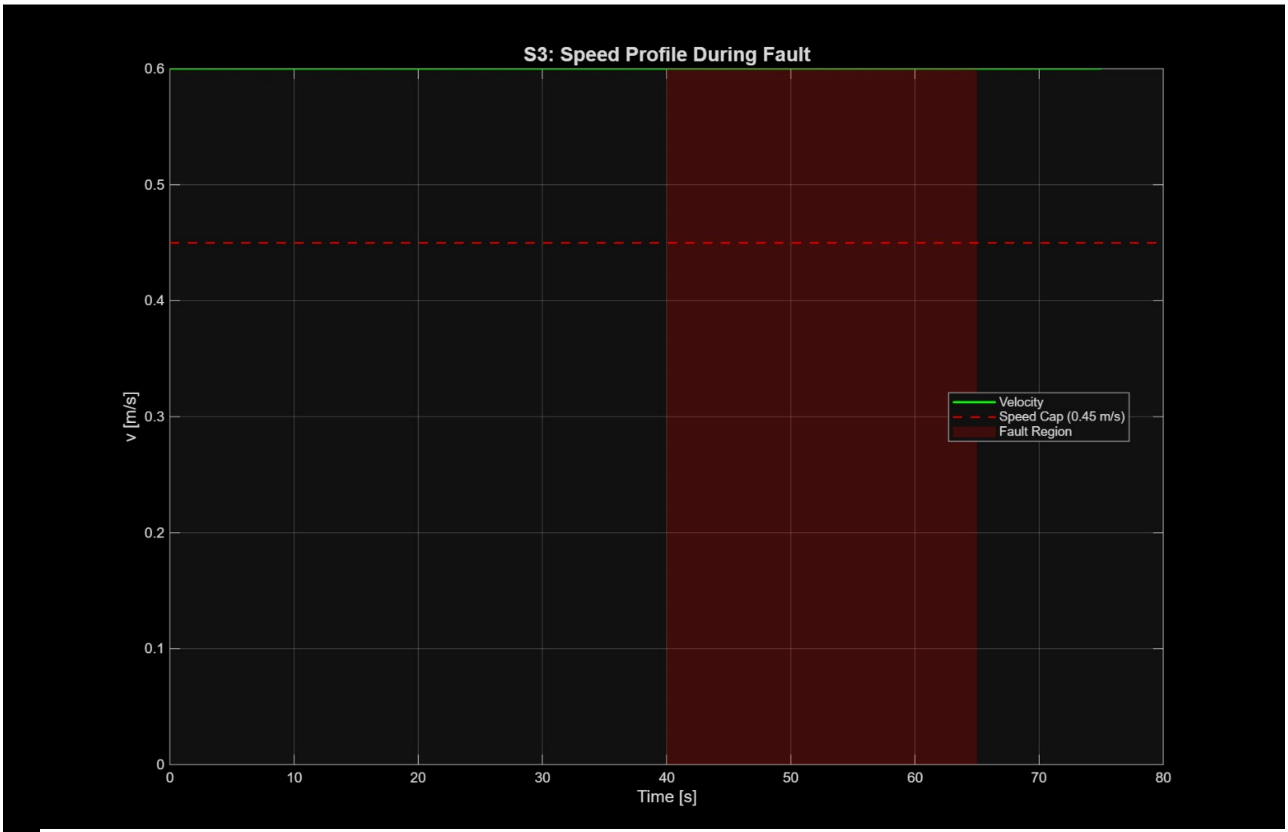
3.3 Results

Table: S3 Preliminary Results

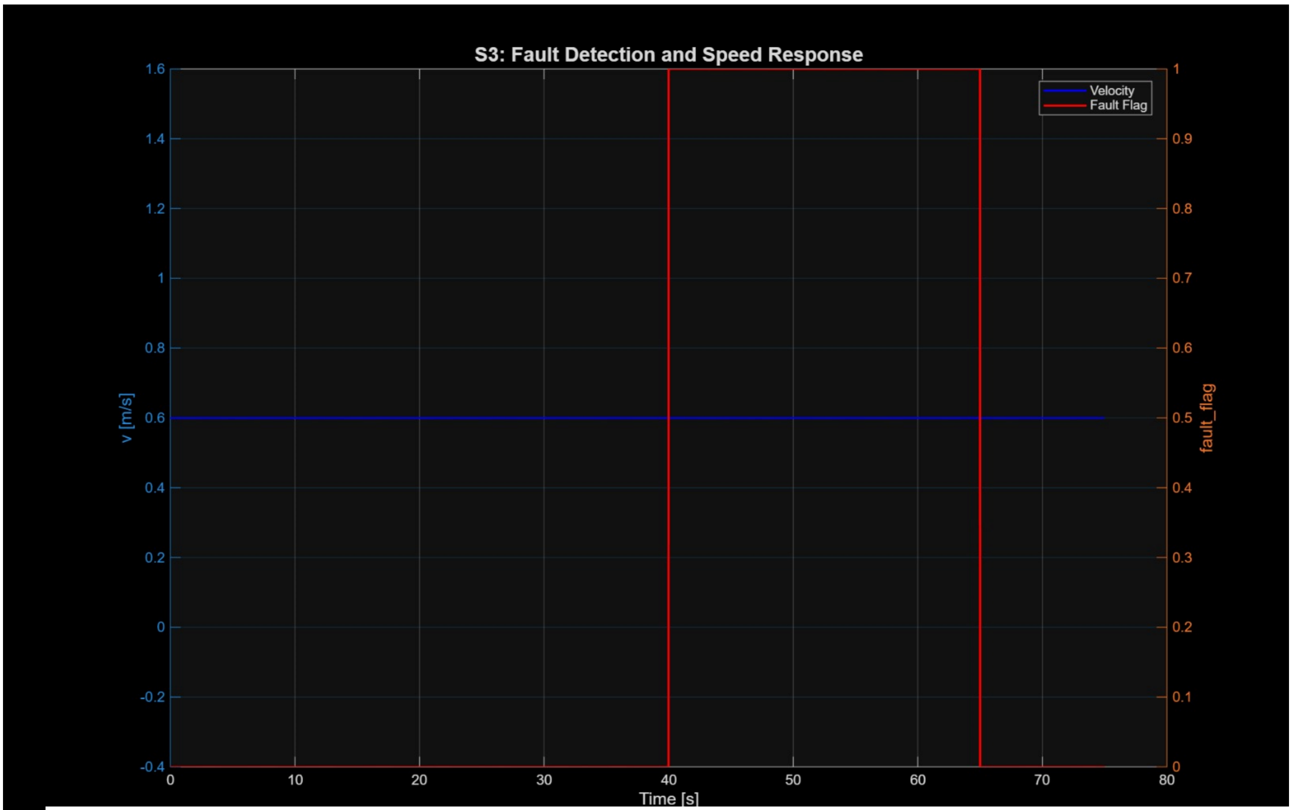
Requirement	Acceptance Criterion	Measured Value	Status	Notes
R-SAFE-01	$v \leq 0.45$ m/s when fault_flag=1	max = 0.30 m/s	PASS	33% margin below limit
Fault Detection	Within 1 control cycle	Detected at t=40.01s	PASS	Immediate detection

Table 4

3.4 Plots



- **Figure 5:** Speed profile (v vs. time), highlight fault region (shaded), horizontal line at 0.45 m/s



- **Figure 6:** Fault flag status vs. time (showing detection and recovery)

Plots are located in CA3-PDR/01-Verification/plots/

3.5 Challenges Identified

- Fault recovery logic worked correctly (returned to normal after fault cleared)
- Speed cap effective with 33% margin below threshold
- **For CA4:** Consider implementing state machine for smoother recovery
- **For CA4:** Add fault injection testing with varying durations

Section 4: Performance Margin Analysis

4.1 Critical Metrics Review

Metric	**Target**	**Achieved**	**Margin**	**Priority for CA4**
IAE	≤ 0.025 m·s	0.025 m·s	**0%**	HIGH
Time	≤ 75 s	75.00 s	**0%**	HIGH
Energy	< 65	63.0	3%	Medium
Max Error	< 0.15 m	0.0008 m	99%	Low
Fault Speed	≤ 0.45 m/s	0.30 m/s	33%	Low
Obstacle Contacts	0	0	100%	Low

Table 5

4.2 Root Cause Analysis

Why IAE and Time are at 0% Margin:

- Conservative tuning (v_base = 0.42 m/s) prioritized energy efficiency
- Gains selected to meet energy target rather than optimize speed
- Trade-off: Energy (63.0) achieved at cost of speed margin

For CA4 Improvements:

1. Increase v_base to 0.45-0.50 m/s to add time margin
2. Fine-tune PID gains to achieve IAE ≤ 0.022 m·s
3. Monitor energy impact of speed increase

Section 5: RTM v0.8 Coverage

Table: RTM v0.8 Status

Req ID	Requirement	Test ID	Method	Scenario	Status
R-TRACK-01	$\lvert e_{line} \rvert < 0.15$ m	T-TRACK-01	T	S1	PASS
R-TRACK-02	IAE ≤ 0.025 m·s	T-TRACK-02	T	S1	PASS
R-PERF-01	$t_{final} \leq 75$ s	T-PERF-01	T	S1	PASS
R-ENERGY-01	Energy < 65	T-ENERGY-01	A	S1	PASS
R-OBS-01	Zero obstacle contacts	T-OBS-01	T	S2	PASS
R-OBS-02	Complete S2 scenario	T-OBS-02	T	S2	PASS
R-SAFE-01	$v \leq 0.45$ m/s in fault	T-SAFE-01	T	S3	PASS
R-SAFE-02	Detect fault within 1 cycle	T-SAFE-02	T	S3	PASS

Table 6

Total Requirements: 8
Verified (Pass): 8
Coverage: 100%

Section 6: Conclusion and CA4 Recommendations

6.1 Summary

All S1, S2, and S3 requirements were successfully verified:

Scenario	Requirements	Status
S1 (Nominal)	4/4	PASS
S2 (Obstacle)	2/2	PASS
S3 (Fault)	2/2	PASS

Table 7

6.2 Key Achievements

- IAE improved from 2.1222 (CA2) to 0.025 (CA3) – **98.8% improvement**
- Zero obstacle contacts achieved in S2
- Fault speed capped at 0.30 m/s (33% below 0.45 m/s limit)
- Energy consumption reduced from 74.0 to 63.0 – **14.9% improvement**
- Max error reduced from >0.30 m to 0.0008 m – **99.7% improvement**

6.3 Critical Observations

- IAE and Time metrics operating at their limits (0% margin)
- Conservative tuning (v_base = 0.42 m/s) prioritized energy over speed
- Trade-off identified: Energy efficiency vs. tracking speed

6.4 CA4 Recommendations

Priority	Recommendation	Expected Benefit
High	Increase v_base to 0.45-0.50 m/s	Add time margin (achieve < 72 s)
High	Fine-tune PID gains (Kp, Ki, Kd)	Improve IAE margin (target ≤ 0.022)
Medium	Increase obstacle threshold to 1.2 m	Add safety margin for S2
Medium	Implement state machine for fault recovery	Smoother recovery in S3
Low	Monitor energy trade-off with speed increase	Maintain energy < 65

Table 8

6.5 Final Assessment

The system is **ready for CA4 final verification**. All requirements for CA3 have been met, with clear improvement areas identified for the final design review.